



HK IMO Training 2022-2023
Problem Set 5 Suggested Solutions



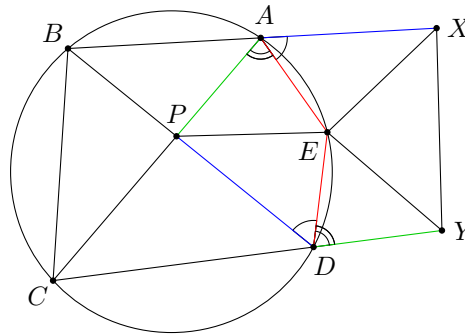
Problem 17 (Dutch MO 2015 Problem 5). Let a, b, c be real numbers satisfying $|a - b| \geq |c|$, $|b - c| \geq |a|$ and $|c - a| \geq |b|$. Prove that one of a, b, c is equal to the sum of the other two.

Solution. Note that $|a - b| \geq |c|$ is the same as $(a - b)^2 \geq c^2$, i.e. $(a - b - c)(a - b + c) \geq 0$. Similarly, we have $(b - c - a)(b - c + a) \geq 0$ and $(c - a - b)(c - a + b) \geq 0$. Multiplying these, we obtain

$$-(a - b - c)^2(b - c - a)^2(c - a - b)^2 \geq 0.$$

Therefore, we must have $(a - b - c)^2(b - c - a)^2(c - a - b)^2 = 0$. This implies $a = b + c$ or $b = c + a$ or $c = a + b$. □

Problem 18 (Canada MO 2018 Problem 2). Let $ABCDE$ be a cyclic pentagon such that $AE = DE$. The diagonals AC and BD meet at P . Let X be a point on the ray BA beyond A such that $AX = PD$. Let Y be a point on the ray CD beyond D such that $DY = PA$. Prove that PE is perpendicular to XY .



Solution. As $AX = DP$, $AE = DE$ and $\angle EAX = \angle EDP$, we have $\triangle EAX \cong \triangle EDP$. This implies $EX = EP$. Similarly, $EY = EP = EX$. Next, as $AP = DY$, $AX = DP$ and $\angle PAX = \angle PAE + \angle EAX = \angle YDE + \angle EDP = \angle YDP$, we have $\triangle PAX \cong \triangle YDP$. This implies $PX = YP$. Therefore, both P and E lie on the perpendicular bisector of XY , which implies $PE \perp XY$. □

Problem 19 (Rioplattense MO 2018 Problem 6). There are n trainees in the IMO training. In a phase, there are 7 classes in total. For every pair of trainees, there are at least 3 classes such that exactly one of them has attended the class (it is not necessary that the same trainee joins all those 3 classes). Also, it is known that one of the trainees has never joined any class in this phase. Find the greatest possible number of n .

Solution. The answer is 16.

We can associate a binary string of 7 digits to each trainee such that a digit 1 means the trainee has attended the corresponding class. For the j th trainee with binary string a_j , let S_j be the set containing all binary strings of 7 digits which differ from a_j in exactly one digit. As every two a_i, a_j differ in at least 3 digits, none of the a_i 's belongs to any S_j , and every two S_i, S_j are disjoint. As $|S_j| = 7$ and there are 2^7 binary strings in total, we have $n + 7n \leq 128$, which gives $n \leq 16$.

An example of $n = 16$ is given below.

0000000, 1110000, 1001100, 1000011, 0101010, 0100101, 0011001, 0010110,
1111111, 0001111, 0110011, 0111100, 1010101, 1011010, 1100110, 1101001

□

Problem 20 (Iran TST 2008 Problem 8). Find all polynomials $P(x)$ with integer coefficients satisfying the following condition: For any positive integers m and n such that $m + n$ is a perfect square, $P(m) + P(n)$ is also a perfect square.

Solution. We first establish two results.

Claim 1: Let $f(x)$ be a polynomial with integer coefficients and even degree such that the leading coefficient is a perfect square. If $f(x)$ is a perfect square for all positive integers x , then $f(x) = g(x)^2$ for some polynomial g with integer coefficients.

Proof. Let $f(x) = c^2x^{2n} + a_{2n-1}x^{2n-1} + \dots + a_1x + a_0$ be such a polynomial. By comparing the coefficients of $x^{2n-1}, \dots, x^n$ one by one, we can define $g(x) = cx^n + b_{n-1}x^{n-1} + \dots + b_1x + b_0 \in \mathbb{Q}[x]$ such that $f(x) - g(x)^2$ is a polynomial of degree at most $n-1$. Let D be the smallest positive integer such that $Dg(x) \in \mathbb{Z}[x]$. Then $D^2f(x) - (Dg(x))^2$ is a polynomial with integer coefficients.

If $D^2f(x) - (Dg(x))^2$ has positive leading coefficient, then $(Dg(x))^2 < D^2f(x) < (Dg(x) + 1)^2$ for all sufficiently large x . This contradicts the assumption that $D^2f(x)$ is a perfect square. Similarly, if $D^2f(x) - (Dg(x))^2$ has negative leading coefficient, then $(Dg(x))^2 > D^2f(x) > (Dg(x) - 1)^2$ for all sufficiently large x , contradiction. Therefore, we must have $D^2f(x) - (Dg(x))^2 = 0$. This implies $f(x) = g(x)^2$, and hence $g \in \mathbb{Z}[x]$ by Gauss's lemma. $\square$

Claim 2: If $f(x)$ is a polynomial with integer coefficients such that $f(x)$ is a perfect square for all positive integers x , then $f(x) = g(x)^2$ for some polynomial g with integer coefficients.

Proof. Clearly, there exists a positive integer m such that the polynomials $f(x)$ and $f(x+m)$ are relatively prime (because this holds as long as they do not share any common complex roots). Note that $f(x)f(x+m)$ is a polynomial with integer coefficients and even degree such that the leading coefficient is a perfect square. Also, $f(x)f(x+m)$ is a perfect square for all positive integers x . By claim 1, $f(x)f(x+m)$ must be the square of a polynomial with integer coefficients. As $f(x)$ and $f(x+m)$ are relatively prime, both of them are squares of some integer polynomials. $\square$

Now, let a, b be fixed positive integers such that $a + b$ is a perfect square. Then $k^2a + k^2b$ is also a perfect square for any positive integer k . Therefore, $P(k^2a) + P(k^2b)$ is a perfect square for all k . By claim 2, $P(k^2a) + P(k^2b) = Q(k)^2$ for some integer polynomial Q depending on a, b . Let

$$P(x) = c_nx^n + c_{n-1}x^{n-1} + \dots + c_0.$$

The leading term of $P(k^2a) + P(k^2b)$ is $c_n(a^n + b^n)k^{2n}$. As it is the leading term of $Q(k)^2$, $c_n(a^n + b^n)$ must be a perfect square.

- If n is even and $n > 0$, we let p be any prime $\equiv 3 \pmod{8}$ which does not divide c_n . Putting $a = 1$ and $b = p^2 + 2p$, we see that $c_n(1 + (p^2 + 2p)^n) \equiv c_n \pmod{p}$ is a quadratic residue modulo p . Putting $a = b = 2$, we see that c_n2^{n+1} is a perfect square. So $2c_n$ is a perfect square, and hence 2 is a quadratic residue modulo p , contradicting $p \equiv 3 \pmod{8}$.
- If $n = 0$, then $P(m) + P(n) = 2c_0$ has to be a perfect square. Therefore, $P(x)$ can be any polynomial $2c^2$ where c is an integer.
- If n is odd, putting $a = b = 2$, we see that c_n2^{n+1} is a perfect square, and hence c_n is a perfect square. Putting $a = 3$ and $b = 1$, we see that $c_n(3^n + 1)$ is a perfect square, and hence $3^n + 1$ is a perfect square. Let $3^n + 1 = d^2$. This gives

$$3^n = (d-1)(d+1).$$

As $\gcd(d-1, d+1) = \gcd(d-1, 2) = 1, 2$, we must have $d-1 = 1$. This yields the only solution $d = 2$ and $n = 1$. Now, $P(m) + P(n) = c_1(m+n) + 2c_0$. This means we want $c_1x^2 + 2c_0$ to be a perfect square for all positive integers x . By claim 2, $c_1x^2 + 2c_0 = (d_1x + d_0)^2$ for some integers d_1, d_0 . But it is obvious that $d_0 = 0$ (as $d_1 \neq 0$), so $P(x) = c^2x$ for some integer c .

Therefore, $P(x) = 2c^2$ and $P(x) = c^2x$ are the only solutions. $\square$